Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. What is output by the following code? (Choose all that apply.)

```
1: public class Fish {
2:    public static void main(String[] args) {
3:       int numFish = 4;
4:       String fishType = "tuna";
5:       String anotherFish = numFish + 1;
6:       System.out.println(anotherFish + " " + fishType);
7:       System.out.println(numFish + " " + 1);
8: } }
```

A. `4 1`

B. `5`

C. `5 tuna` No se puede almacenar un int en una variable de tipo String

D. `5tuna`

E. `51tuna`

F. The code does not compile.

2. Which of these array declarations are not legal? (Choose all that apply.)

A. `int[][] scores = new int[5][];`

B. `Object[][][] cubbies = new Object[3][0][5];`

C. `String beans[] = new beans[6];` No se puede usar un nombre de variable como tipo

D. `java.util.Date[] dates[] = new java.util.Date[2][];`

E. `int[][] types = new int[];` No se especifica el tamaño del arreglo.

F. `int[][] java = new int[][];`

3. Note that March 13, 2022 is the weekend when we spring forward, and November 6, 2022 is when we fall back for daylight saving time. Which of the following can fill in the blank without the code throwing an exception? (Choose all that apply.)

```
var zone = ZoneId.of("US/Eastern");
var date = ______________________________;
var time = LocalTime.of(2, 15);
var z = ZonedDateTime.of(date, time, zone);
```

A. `LocalDate.of(2022, 3, 13)` Fecha válida

B. `LocalDate.of(2022, 3, 40)`

C. `LocalDate.of(2022, 11, 6)` Fecha válida

D. `LocalDate.of(2022, 11, 7)` Fecha válida

E. `LocalDate.of(2023, 2, 29)`

F. `LocalDate.of(2022, MonthEnum.MARCH, 13);`

4. Which of the following are output by this code? (Choose all that apply.)

```
3: var s = "Hello";
4: var t = new String(s);
5: if ("Hello".equals(s)) System.out.println("one");
6: if (t == s) System.out.println("two");
7: if (t.intern() == s) System.out.println("three");
8: if ("Hello" == s) System.out.println("four");
9: if ("Hello".intern() == t) System.out.println("five");
```

A. `one` Compara las cadenas de texto "Hello"

B. `two`

C. `three` El método intern() busca la cadena en el String Pool. Si ya existe devuelve la referencia de esa cadena en el pool.

D. `four` "Hello" y s apuntan exactamente al mismo objeto dentro del String Pool

E. `five`

F. The code does not compile.

G. None of the above

5. What is the result of the following code?

```
7: var sb = new StringBuilder();
8: sb.append("aaa").insert(1, "bb").insert(4, "ccc");
9: System.out.println(sb);
```

A. `abbaaccc`

B. `abbaccca`

C. `bbaaaccc`

D. `bbaaccca`

E. An empty line

F. The code does not compile.

6. How many of these lines contain a compiler error? (Choose all that apply.)

```
23: double one = Math.pow(1, 2);
24: int two = Math.round(1.0);
25: float three = Math.random();
26: var doubles = new double[] {one, two, three};
```

Math.Round devuelve un float cuando el parámetro es flotante.

Math.random() retorna un double, pero three espera un float.

A. 0

B. 1

C. 2

D. 3

E. 4

7. Which of these statements is true of the two values? (Choose all that apply.)

```
2022-08-28T05:00 GMT-04:00
2022-08-28T09:00 GMT-06:00
```

5 + 4 = 9, esta es la hora con respecto al UTC (Tiempo universal coordinado)

6+9 = 15

A. The first date/time is earlier. Sí 9 > 15 hrs

B. The second date/time is earlier.

C. Both date/times are the same.

D. The date/times are two hours apart.

E. The date/times are six hours apart. 15 - 9 = 6

F. The date/times are 10 hours apart.

8. Which of the following return 5 when run independently? (Choose all that apply.)

```
var string = "12345";
var builder = new StringBuilder("12345");
```

A. `builder.charAt(4)` 1 2 3 4 5 / 0-1-2-3-4 Devuelve 5

B. `builder.replace(2, 4, "6").charAt(3)` "12345" -> "1265" -> charat(3) = 5

C. `builder.replace(2, 5, "6").charAt(2)`

D. `string.charAt(5)`

E. `string.length`

F. `string.replace("123", "1").charAt(2)` "12345" -> "145" -> 5

G. None of the above

9. Which of the following are true about arrays? (Choose all that apply.)

A. The first element is index 0.

B. The first element is index 1.

C. Arrays are fixed size.

D. Arrays are immutable.

E. Calling `equals()` on two different arrays containing the same primitive values always returns `true`.

F. Calling `equals()` on two different arrays containing the same primitive values always returns `false`.

G. Calling `equals()` on two different arrays containing the same primitive values can return `true` or `false`.

10. How many of these lines contain a compiler error? (Choose all that apply.)

```
23: int one = Math.min(5, 3);
24: long two = Math.round(5.5);
25: double three = Math.floor(6.6);
26: var doubles = new double[] {one, two, three};
```

A. 0
B. 1
C. 2
D. 3
E. 4

11. What is the output of the following code?

```
var date = LocalDate.of(2022, 4, 3);
date.plusDays(2);
date.plusHours(3);
System.out.println(date.getYear() + " " + date.getMonth()
    + " " + date.getDayOfMonth());
```

A. `2022 MARCH 4`
B. `2022 MARCH 6`
C. `2022 APRIL 3`
D. `2022 APRIL 5`
E. The code does not compile.
F. A runtime exception is thrown.

12. What is output by the following code? (Choose all that apply.)

```
var numbers = "012345678".indent(1);
numbers = numbers.stripLeading();
System.out.println(numbers.substring(1, 3));
System.out.println(numbers.substring(7, 7));
System.out.print(numbers.substring(7));
```

A. `12`
B. `123`
C. `7`
D. `78`
E. A blank line
F. The code does not compile.
G. An exception is thrown.

12
(espacio)
78

13. What is the result of the following code?

```
public class Lion {
   public void roar(String roar1, StringBuilder roar2) {
      roar1.concat("!!!");
      roar2.append("!!!");
   }
   public static void main(String[] args) {
      var roar1 = "roar";
      var roar2 = new StringBuilder("roar");
      new Lion().roar(roar1, roar2);
      System.out.println(roar1 + " " + roar2);
} }
```

A. `roar roar`

B. `roar roar!!!`

C. `roar!!! roar`

D. `roar!!! roar!!!`

E. An exception is thrown.

F. The code does not compile.

14. Given the following, which can correctly fill in the blank? (Choose all that apply.)

```
var date = LocalDate.now();
var time = LocalTime.now();
var dateTime = LocalDateTime.now();
var zoneId = ZoneId.systemDefault();
var zonedDateTime = ZonedDateTime.of(dateTime, zoneId);
Instant instant = ______________________________;
```

A. `Instant.now()` Método por defecto que pregunta al reloj del sistema el momento justo

B. `new Instant()`

C. `date.toInstant()`

D. `dateTime.toInstant()`

E. `time.toInstant()`

F. `zonedDateTime.toInstant()` Java calcula el offset exacto entre UTC y GMT

15. What is the output of the following? (Choose all that apply.)

```
var arr = new String[] { "PIG", "pig", "123"};
Arrays.sort(arr);
System.out.println(Arrays.toString(arr));
System.out.println(Arrays.binarySearch(arr, "Pippa"));
```

A. `[pig, PIG, 123]`

B. `[PIG, pig, 123]`

C. `[123, PIG, pig]` El órden está dado por el código UNICODE, números, mayusculas y minúsculas,

D. `[123, pig, PIG]`

E. `-3` Como java no encuentra "Pippa" calcula el punto de inserción resultando en -(2)-1 = -3

F. `-2`

G. The results of `binarySearch()` are undefined in this example.

16. What is included in the output of the following code? (Choose all that apply.)

```
var base = "ewe\nsheep\\t";
int length = base.length();
int indent = base.indent(2).length();
int translate = base.translateEscapes().length();

var formatted = "%s %s %s".formatted(length, indent, translate);
System.out.format(formatted);
```

A. `10` base.traslateEscapes().length

B. `11` base.length

C. `12`

D. `13`

E. `14`

F. `15`

G. `16` base.intend(2).length()

17. Which of these statements are true? (Choose all that apply.)

```
var letters = new StringBuilder("abcdefg");
```

A. `letters.substring(1, 2)` returns a single-character `String`. Solo devolverá un character

B. `letters.substring(2, 2)` returns a single-character `String`.

C. `letters.substring(6, 5)` returns a single-character `String`.

D. `letters.substring(6, 6)` returns a single-character `String`.

E. `letters.substring(1, 2)` throws an exception.

F. `letters.substring(2, 2)` throws an exception.

G. `letters.substring(6, 5)` throws an exception. No podemos iniciar con un número mayor y terminar en uno menor

H. `letters.substring(6, 6)` throws an exception.

18. What is the result of the following code? (Choose all that apply.)

```
13: String s1 = """
14:    purr""";
```

```
15: String s2 = "";
16:
17: s1.toUpperCase();
18: s1.trim();
19: s1.substring(1, 3);
20: s1 += "two"; "purrtwo"
21:
22: s2 += 2;
23: s2 += 'c';
24: s2 += false;
25:
26: if ( s2 == "2cfalse") System.out.println("==");
27: if ( s2.equals("2cfalse")) System.out.println("equals");
28: System.out.println(s1.length());
```

A. 2

B. 4

C. 7 "purrtwo"

D. 10

E. ==

F. equals No comparamos memoria, sino contenido: if (s2.equals("2cfalse")) System.out.println("equals");

G. An exception is thrown.

H. The code does not compile.

19. Which of the following fill in the blank to print a positive integer? (Choose all that apply.)

```
String[] s1 = { "Camel", "Peacock", "Llama"};
String[] s2 = { "Camel", "Llama", "Peacock"};
String[] s3 = { "Camel"};
String[] s4 = { "Camel", null};
System.out.println(Arrays._______________________);
```

A. compare(s1, s2) En el índice 1, s1 tiene "Peacock" y s2 tiene "Llama", 'Peacock' es mayor que "Llama"

B. mismatch(s1, s2) La primera discrepancia ocurre en el índice 1, entonces se devuelve 1

C. compare(s3, s4)

D. mismatch (s3, s4) S3 ya no tiene elementos en el indice 1 y S4 tiene null, devuelve 1.

E. compare(s4, s4)

F. mismatch (s4, s4)

20. Note that March 13, 2022 is the weekend that clocks spring ahead for daylight saving time. What is the output of the following? (Choose all that apply.)

```
var date = LocalDate.of(2022, Month.MARCH, 13);
var time = LocalTime.of(1, 30);
var zone = ZoneId.of("US/Eastern");
var dateTime1 = ZonedDateTime.of(date, time, zone);
var dateTime2 = dateTime1.plus(1, ChronoUnit.HOURS);

long diff = ChronoUnit.HOURS.between(dateTime1, dateTime2);
int hour = dateTime2.getHour();
boolean offset = dateTime1.getOffset()
   == dateTime2.getOffset();
System.out.println("diff = " + diff);
System.out.println("hour = " + hour);
System.out.println("offset = " + offset);
```

A. `diff = 1` between() mide el tiempo real y como se le sumó 1 con plus la diferencia sigue siendo una hora.

B. `diff = 2`

C. `hour = 2`

D. `hour = 3` getHour da la hora local, al sumarle 1 hora real el reloj salta a las 3:30.

E. `offset = true`

F. The code does not compile.

G. A runtime exception is thrown.

21. Which of the following can fill in the blank to print `avaJ`? (Choose all that apply.)

```
3: var puzzle = new StringBuilder("Java");
4: puzzle._____;
5: System.out.println(puzzle);
```

A. `reverse()` Método diseñado para invertir el contenido

B. `append("vaJ$").substring(0, 4)` append("vaj$") añade el texto al final

C. `append("vaJ$").delete(0, 3).deleteCharAt(puzzle.length() - 1)` se eliminan los primeros caracteres

D. `append("vaJ$").delete(0, 3).deleteCharAt(puzzle.length())` se elimina el último caracter

E. None of the above

22. What is the output of the following code?

```
var date = LocalDate.of(2022, Month.APRIL, 30);
date.plusDays(2);
date.plusYears(3);
System.out.println(date.getYear() + " " + date.getMonth()
   + " " + date.getDayOfMonth());
```

A. `2022 APRIL 30` LocalDate es inmutable

B. `2022 MAY 2`

C. `2025 APRIL 2`

D. `2025 APRIL 30`

E. `2025 MAY 2`

F. The code does not compile.

G. A runtime exception is thrown.